

Note on: *Hedging the Treasury Lock*

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Reading notes

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One-paragraph summary

A T-Lock pays the client $N \cdot \text{RiskFactor}_{t_e} \cdot (\text{IRR}_{t_e} - L)$ at expiry t_e , where IRR is the bond's internal rate of return and $\text{RiskFactor} = -\partial P / \partial y \cdot 10^4$ (i.e. $-10^4 \cdot \text{DV01}$). Because $P(y)$ is convex, the linear payoff in IRR is, in fact, *globally below* the negative of P when viewed as a function of yield — so booking a short T-Lock as a long forward contract on the same bond, struck at $P_{t_e}(L)$, is an overhedge. The paper quantifies that overhedge error, shows the gamma is locally negative (so delta-hedging the proxy forward generates gamma profits), and extends the argument to Then-Current-T-Locks via a benchmark-roll P&L analysis backed by 2008–2019 Treasury auction statistics. The natural hedge is a reverse repo on the underlying bond; the bond-specific repo rate is the right drift in the forward price.

1 Setup and notation

The paper's symbols mostly align with the house notation, with one collision worth flagging: $P_t(y)$ in the paper is the *bond* price as a function of yield, not a discount factor. House notation keeps $P_t(y)$ for the bond and $D_{t,T}$ for the discount factor; they should not be confused.

Paper	House	Meaning
$P_t(y)$	$P_t(y)$	Dirty price of the bond at time t as a function of yield y
P_t^{mkt}	$P_t(\text{IRR}_t)$	Observed market dirty price
IRR_t	IRR_t	Internal rate of return solving $P_t(\text{IRR}_t) = P_t^{mkt}$
L	L	Locked yield (strike)
RiskFactor_t	RiskFactor_t	$-\partial P_t(y)/\partial y$ at $y = \text{IRR}_t$. Equals $-10^4 \cdot \text{DV01}$.
D_{t,t_e}	D_{t,t_e}	Risk-free DF from t to t_e
$\text{ForwardPrice}_t(t_e)$	$\text{Fwd}_t(t_e)$	Forward bond price for delivery at t_e
r_{t,t_e}^{repo}	r_{t,t_e}^{repo}	Simply-compounded forward repo rate $t \rightarrow t_e$
a	$a \in \{+1, -1\}$	Long (+1) / short (-1) T-Lock
$P_t^{(0)}$	—	Bond price stripped of coupons ($c = 0$); zero-bond proxy
$A_t(y)$	$A_t(y)$	Coupon annuity $\sum \alpha_i e^{-(t_i-t)y}$

2 Key equations

Payoff and IRR definition.

$$g = a \cdot \text{RiskFactor}_{t_e} \cdot (\text{IRR}_{t_e} - L). \quad (1)$$

Says: a long ($a = 1$) T-Lock pays the bondholder when yields rise (because $\text{RiskFactor} > 0$); a short pays when yields fall.

$$P_t(\text{IRR}_t) = P_t^{mkt}, \quad P_t(y) = \prod_{i=1}^n \frac{1}{1 + \alpha_i y} + c \sum_{i=1}^n \alpha_i \prod_{j=1}^i \frac{1}{1 + \alpha_j y}. \quad (2)$$

Says: the IRR is the yield that reprices the bond to its dirty market price; $P_t(\cdot)$ uses the *compounded* yield convention (not continuously compounded — this matters for code).

Continuous-compounding form (used for greeks). For the sensitivity analysis, the paper switches to

$$P_t(y) = e^{-(t_n-t)y} + c \sum_{t_i > t} \alpha_i e^{-(t_i-t)y}, \quad (3)$$

with derivatives (B.84–87)

$$D_y^k [P_t(\cdot)] = (t - t_n)^k e^{-(t_n-t)y} + c \sum_{t_i > t} \alpha_i (t - t_i)^k e^{-(t_i-t)y}. \quad (4)$$

Says: closed-form derivatives of all orders in y , sign alternating with k . $D_y^1 < 0$, $D_y^2 > 0$, $D_y^3 < 0$.

Linear (forward) approximation.

$$g \simeq a \cdot (P_{t_e}(L) - P_{t_e}(\text{IRR}_{t_e})) = a \cdot (P_{t_e}(L) - P_{t_e}^{mkt}). \quad (5)$$

Says: at first order, the T-Lock is a forward contract on the bond struck at $K = P_{t_e}(L)$. Booking a short T-Lock as a long forward at this strike is the operational shortcut.

Overhedge error bound. With the remainder $R_1(y) = P(y) - P(L) - D_y^1[P(\cdot)](y - L)$,

$$|R_1(y)| \leq \frac{M(y - L)^2}{2}, \quad M = \max_I |D_y^2[P(\cdot)]|. \quad (6)$$

Says: for the 10Y benchmark with $M \approx 150$ and a 30% yield change (i.e. $|y - L| \approx 0.3L$), the price error is about **0.005%** on 100% notional. The approximation is tight.

Delta and gamma.

$$\Delta = D_P^1[g] \cdot D_{t,t_e}, \quad D_y^1[g] = -a \cdot [D_y^2[P(\cdot)](y - L) + D_y^1[P(\cdot)]], \quad (7)$$

$$\Gamma = D_P^2[g] \simeq -a \cdot D_y^2[P(\cdot)] \cdot [1/D_y^1[P(\cdot)]]^2 \quad (\text{local, } |y - L| \text{ small}). \quad (8)$$

Says: for $a = 1$ near L , $\Delta < 0$ (consistent with “long T-Lock = synthetic short bond”) and $\Gamma < 0$. The negative gamma is what makes the linear proxy an overhedge — the bank gains on dynamic delta-rebalancing.

Sharp sign conditions (resolve (7) and the gamma directly):

$$\Delta < 0 \iff y - L < -D_y^1[P(\cdot)] / D_y^2[P(\cdot)] > 0, \quad (9)$$

$$\Gamma < 0 \iff y - L < \frac{D_y^1[P(\cdot)] \cdot D_y^3[P(\cdot)]}{(D_y^2[P(\cdot)])^2 - D_y^1[P(\cdot)] \cdot D_y^3[P(\cdot)]} > 0. \quad (10)$$

Says: both bounds give exact yield thresholds beyond which the sign flips. In “normal” yield regimes (y not far above L), $\Delta < 0$ and $\Gamma < 0$ both hold.

Forward price via repo (Appendix D). For zero haircut,

$$\text{Fwd}_t(t_e) = P_t^{\text{mkt}} e^{r^{\text{repo}} \cdot (t_e - t)} - \sum_{t < t_i \leq t_e} c_i e^{r^{\text{repo}} \cdot (t_e - t_i)}. \quad (11)$$

Says: the forward bond price drifts at the *repo* rate, not the OIS / risk-free rate. Coupons paid before t_e are forwarded to t_e at the same repo rate. With a haircut h blend in the unsecured funding rate r^{fun} .

Then-Current-T-Lock: strike at roll.

$$K = 1 + \text{RiskFactor}_{t_e} \cdot (R_t - L), \quad (12)$$

$$\text{StrikeGap} = (\widehat{\text{RiskFactor}}_{t_e} - \text{RiskFactor}_{t_e}) \cdot (\hat{R}_{t_{\text{roll}}} - L) - \text{RiskFactor}_{t_e} \cdot (R_{t_{\text{roll}}} - \hat{R}_{t_{\text{roll}}}). \quad (13)$$

Says: when the on-the-run rolls, the strike updates. The first term captures the risk-factor jump between old and new benchmark; the second the IRR jump at the roll date.

3 Derivations & commentary

Commentary

The key trick is (5): turning the T-Lock payoff $g = -\Delta P(\cdot) \cdot (y - L)$ into the secant $P_{t_e}(L) - P_{t_e}(\text{IRR}_{t_e})$ uses the first-order Taylor identity $P(\text{IRR}) - P(L) \simeq D_y^1[P(\cdot)](\text{IRR} - L)$. The sign of the remainder is fixed by convexity ($D_y^2[P(\cdot)] > 0$ always for a bond with positive cashflows), which is why the bank’s position is conservative *globally*, not just locally. Importantly, the gamma being negative is the *opposite* sign to what a naive “T-Lock $\approx$ CMS-let” intuition would give — the CMT interpretation freezes RiskFactor_{t_e} and obscures the dependence of the risk factor on yield (which is what flips the convexity).

Commentary

The repo rate in (11) is bond-specific. A benchmark Treasury can trade “special” — the paper notes historical episodes of the 10Y repo rate going to -300 bps. For validation, the test case in the paper uses GC rates (Tab. 1); the specialness premium must be added

as a separate spread input.

Commentary

The continuous-compounding form (3) is used purely for the analytic greeks. The *contract definition* (2) uses compounded (i.e. $1/(1 + \alpha y)$) discounting — mixing them in a pricing function is a common bug. House rule: keep two separate bond-price functions, one per convention, and only the compounded one defines IRR.

4 Validation

Validation

This is a *specification* of mathematical objects the paper defines and the numerical values they should take. It says nothing about how those objects are implemented or invoked.

Quantities the paper defines:

- $P_t(y)$, the bond dirty price as a function of yield, in two distinct conventions: *compounded* (eq. 2) and *continuously compounded* (eq. 3). Only the compounded form defines the contractual IRR.
- IRR_t , the unique solution of $P_t(\text{IRR}_t) = P_t^{\text{mkt}}$ with $P_t(\cdot)$ in the compounded convention.
- $D_y^k[P_t(\cdot)]$, the k -th derivative of the bond price with respect to yield, in closed form (eq. 4).
- $\text{RiskFactor}_t = -D_y^1[P_t(y)]|_{y=\text{IRR}_t}$.
- $\text{Fwd}_t(t_e)$, the t_e -forward bond price under a repo rate (eq. 11), zero-haircut limit.
- $g(y)$, the T-Lock payoff as a function of yield (eq. 1), and its derivatives $D_y^1[g]$, $D_y^2[g]$ in closed form (eqs. 7, 8).

Canonical numerical case (paper, p. 4):

- Instrument: US Treasury, coupon 3.125%, maturity 25-Nov-2028, ISIN US9128285M81.
- As-of: 24 January 2019; $t_e - t = 3$ months.
- $P^{\text{mkt}} = 104.1055$ (dirty), clean price = 103.4922.
- $\text{IRR} = 2.717\%$ spot.
- 3M GC repo $r^{\text{repo}} = 2.46\%$ (mid of bid/ask, Tab. 1).
- Forward: $\text{IRR}_{\text{fwd}} = 2.53\%$, dirty 104.74%, clean 103.36%.
- $L = 2.717\%$.

Expected numerical values:

Quantity	Value	Tolerance
$\text{Fwd}_t(t_e)$ (dirty)	104.74	± 0.01
$\text{Fwd}_t(t_e)$ (clean)	103.36	± 0.01
Forward IRR	2.53%	± 1 bp
$ R_1(y) $ at 10Y, $ y - L \approx 0.3L$	$\sim 0.005\%$ notional	order-of-mag.
$M = \max_I D_y^2[P(\cdot)] $ (10Y benchmark)	~ 150	order-of-mag.

Repo rates for the 10Y as of 24-Jan-2019 (Paper Tab. 1)

Maturity	bid	ask
1M	2.57%	2.52%
3M	2.60%	2.55%
6M	2.63%	2.58%
12M	2.70%	2.65%

Equation $\leftrightarrow$ quantity map:

Paper eq.	Symbol	Quantity
(2)	$P_t(y), \text{IRR}_t$	dirty price (compounded); the y for which $P_t(y) = P_t^{mkt}$
(3)	$P_t(y)$ (CC)	dirty price (continuously compounded)
(4)	$D_y^k[P(\cdot)]$	closed-form k -th yield derivative
(5)	$g \simeq a(P_{t_e}(L) - P_{t_e}^{mkt})$	first-order proxy for the T-Lock payoff
(11)	$\text{Fwd}_t(t_e)$	forward bond price under repo (zero haircut)
(7),(8)	Δ, Γ	T-Lock first and second derivatives

Invariants and identities to verify:

1. $P_t(\text{IRR}_t) = P_t^{mkt}$ holds at the canonical case ($P^{mkt} = 104.1055$, $\text{IRR} = 2.717\%$) within float tolerance.
2. Eq. (11) evaluated at the canonical inputs equals 104.74 (dirty) within ± 0.01 .
3. *Overhedge inequality*: for any L , any y in a reasonable yield range, and any positive-coupon schedule,

$$P(L) + D_y^1[P(\cdot)](y - L) \leq P(y).$$

A global property, following from $D_y^2[P(\cdot)] > 0$.

4. *Gamma sign (local)*: for $a = 1$ and y in a neighbourhood of L , $D_y^2[g] < 0$. Sharp threshold given by (10).
5. *Closed-form vs. numerical agreement*: the closed-form derivative (4) equals a numerical derivative of $P_t(y)$ at any test yield.
6. *Monotonicity in repo*: $\text{Fwd}_t(t_e)$ in (11) is strictly increasing in r^{repo} (under the standing assumption that the spot price exceeds the present value of intermediate coupons — true here).

5 Glossary of symbols (paper's notation)

$P_t(y)$	Bond dirty price at t as a function of yield y
P_t^{mkt}	Observed market dirty price ($= P_t(\text{IRR}_t)$)
IRR_t	Internal rate of return at t
L	Locked yield (T-Lock strike)
RiskFactor_t	$-\partial P_t / \partial y$ at $y = \text{IRR}_t$; equivalently $-10^4 \cdot \text{DV01}$
DV01	Price change per 1 bp positive yield shift
g	T-Lock payoff at expiry
a	+1 (long) or -1 (short) unitary T-Lock
t_e	T-Lock expiry
D_{tt_e}	Risk-free discount factor from t to t_e
$D_y^k[\cdot]$	Euler shorthand for $\partial^k / \partial y^k$
$P_t^{(0)}$	Bond price stripped of coupons (coupon $c = 0$); pure-discount component
$A_t(y)$	Coupon annuity $\sum_{t_i > t} \alpha_i e^{-(t_i - t)y}$ (CC form)
α_i	Year fraction for accrual period ending at t_i
r_{uv}^{repo}	Simply-compounded forward repo rate, $u \rightarrow v$
r^{fun}	Unsecured funding rate
h^{cut}	Repo haircut, $\in [0, 1]$
$\text{ForwardPrice}_t(t_e)$	t_e -forward bond price at t
R_t ($\hat{R}_t$)	IRR of the current (new) on-the-run benchmark at t
t_{roll}	Date of benchmark roll in a Then-Current-T-Lock

TODO / Open questions

- Reproduce Fig. 1 (T-Lock payoff vs. forward overhedge) on the 24-Jan-2019 test case. Provides a visual smoke test for the dirty-price and forward-price formulas.
- Reproduce Fig. 2 & 3 (Delta, Gamma plots) — shows the sign-flip at extreme yields predicted by (9) and (10).
- Section 4 statistical analysis: implement the roll-P&L estimator and check against paper's Table 2 (IRR statistics from 2008 Treasury auction history). Lower priority since this is empirical, not formula-validation.
- Repo-special scenario: re-price the test case with r^{repo} shifted -300 bps (paper p. 12) to confirm forward price moves in the documented direction.
- Open question: paper assumes a flat short-term repo curve in Sec. 2.4 (around eq. 50). How sensitive is the analysis to the curve shape? Should the validation suite include a non-flat repo curve case?